ANSWERS

Chapter 1 Similarity

Practice set 1.1

1. $\frac{3}{4}$ **2.** $\frac{1}{2}$ **3.** 3 **4.** 1:1 **5.** (1) $\frac{BQ}{BC}$, (2) $\frac{PQ}{AD}$, (3) $\frac{BC}{DC}$, (4) $\frac{DC \times AD}{QC \times PQ}$

Practice set 1.2

1. (1) is a bisector. (2) is not a bisector. (3) is a bisector.

2. $\frac{PN}{NR} = \frac{PM}{MQ} = \frac{3}{2}$, therefore line NM || side RQ **3.** QP = 3.5 **5.** BQ = 17.5

6. QP = 22.4 **7.** $x = 6$; AE = 18 **8.** LT = 4.8 **9.** $x = 10$

10. Given, XQ, PD, Given, $\frac{\boxed{XR}}{\boxed{RF}} = \frac{\boxed{XQ}}{\boxed{QE}}$, Basic propotionality theorem, $\frac{\boxed{XP}}{\boxed{PD}} = \frac{\boxed{XR}}{\boxed{RF}}$

Practice set 1.3

1. Δ ABC ~ Δ EDC, AA test **2.** Δ PQR ~ Δ LMN ; SSS test of similarity

3. 12 metre **4.** AC = 10.5 **6.** OD = 4.5

Practice set 1.4

1. Ratio of areas = 9 : 25 **2.** $\boxed{PQ^2}$, $\frac{\boxed{4}}{\boxed{9}}$ **3.** $\boxed{A(\Delta PQR)}$, $\frac{16}{25}$, $\frac{4}{5}$

4. MN = 15 **5.** 20 cm **6.** $4\sqrt{2}$

7. $\boxed{PF}$; $\boxed{x} + \boxed{2x}$; $\boxed{\angle FPQ}$; $\boxed{\angle FQP}$; $\frac{DF^2}{PF^2}$; $\boxed{20}$; $\boxed{45}$; $\boxed{45} - \boxed{20}$; $\boxed{25 \text{ sq. unit}}$

Problem set 1

1. (1) (B), (2) (B), (3) (B), (4) (D), (5) (A)

2. $\frac{7}{13}$, $\frac{7}{20}$, $\frac{13}{20}$ **3.** 9 cm **4.** $\frac{3}{4}$ **5.** 11 cm **6.** $\frac{25}{81}$ **7.** 4

8. PQ = 80, QR = $\frac{280}{3}$, RS = $\frac{320}{3}$ **9.** $\frac{\boxed{PM}}{\boxed{MQ}} = \frac{\boxed{PX}}{\boxed{XQ}}$, $\frac{\boxed{PM}}{\boxed{MR}} = \frac{\boxed{PY}}{\boxed{YR}}$,

10. $\frac{AX}{XY} = \frac{3}{2}$ **12.** $\frac{\boxed{3}}{\boxed{2}}$, $\frac{\boxed{3}+\boxed{2}}{\boxed{2}}$, $\frac{\boxed{5}}{\boxed{3}}$, $\boxed{AA}$, $\frac{\boxed{5}}{\boxed{3}}$, $\boxed{15}$

Chapter 2 Pythagoras Theorem

Practice set 2.1

1. Pythagorean triplets ; (1), (3), (4), (6) **2.** NQ = 6 **3.** QR = 20.5

4. RP = 12, PS = $6\sqrt{3}$

5. $\boxed{\text{Given}}$, $\boxed{45^\circ}$, $\boxed{\frac{1}{\sqrt{2}}}$, $\boxed{\frac{1}{\sqrt{2}}}$, $\boxed{\frac{1}{\sqrt{2}}}$, $\boxed{2}$

6. side = $5\sqrt{2}$ cm, perimeter = $20\sqrt{2}$ cm **7.** (1) 18 (2) $4\sqrt{13}$ (3) $6\sqrt{13}$ **8.** 37 cm
10. 8.2 metre.

Practice set 2.2

1. 12 **2.** $2\sqrt{10}$ **4.** 18 cm

Problem set 2

1. (1) (B), (2) (B), (3) (A), (4) (C), (5) (D), (6) (C), (7) (B), (8) (A).
2. (1) $a\sqrt{3}$, (2) form a right angled triangle. (3) 61 cm, (4) 15 cm, (5) $x\sqrt{2}$, (6) $\angle$ PRQ.
3. RS = 6 cm, ST = $6\sqrt{3}$ cm **4.** 20 cm **5.** side = 2 cm, perimeter = 6 cm
6. 7 **7.** AP = $2\sqrt{7}$ cm **10.** 7.5 km / hr **12.** 8 cm **14.** 8 cm
15. 192 sq.unit **17.** 58 **18.** 26

Chapter 3 Circle

Practice set 3.1

1. (1) 90°, tangent-radius theorem (2) 6 cm ; perpendicular distance
(3) $6\sqrt{2}$ cm (4) 45°
2. (1) $5\sqrt{3}$ cm (2) 30° (3) 60° **4.** 9 cm

Practice set 3.2

1. 1.3 cm 2. 9.7 cm **4.** (3) 110° **5.** $4\sqrt{6}$ cm

Practice set 3.3

1. m(arc DE) = 90°, m(arc DEF) = 160°

Practice set 3.4

1. (1) 60° (2) 30° (3) 60° (4) 300° **2.** (1) 70° (2) 220° (3) 110° (4) 55°
3. $\angle R = 92^\circ$; $\angle N = 88^\circ$ **7.** 44° **8.** 121°

Practice set 3.5

1. PS = 18; RS = 10, **2.** (1) 7.5 (2) 12 or 6
3. (1) 18 (2) 10 (3) 5 **4.** 4

Problem set 3

1. (1) D (2) B (3) B (4) C (5) B (6) D (7) A (8) B (9) A (10) C.
2. (1) 9 cm (2) in the interior of the circle (3) 2 locations, 12 cm
3. (1) 6 (2) $\angle K = 30^\circ$; $\angle M = 60^\circ$ **5.** 10 **6.** (1) 9 cm (2) 6.5 cm

(3) 90° ; MS : SR = 2 : 1 **9.** $4\sqrt{3}$ cm

13. (1) 180° (2) $\angle$ AQP $\cong$ $\angle$ ASQ $\cong$ $\angle$ ATQ

(3) $\angle$ QTS $\cong$ $\angle$ SQR $\cong$ $\angle$ SAQ (4) 65°, 130° (5) 100° **14.**(1) 70°

(2) 130° (3) 210° **15.** (1) 56° (2) 6 (3) 16 or 9 **16.** (1) 15.5°

(2) 3.36 (3) 6 **18.** (1) 68° (2) OR = 16.2, QR = 13 (3) 13 **21.** 13

Chapter 4 Geometric Constructions

Problem set 4

1. (1) C (2) A (3) A

Chapter 5 Co-ordinate Geometry

Practice set 5.1

1. (1) $2\sqrt{2}$ (2) $4\sqrt{2}$ (3) $\frac{11}{2}$ (4) 13 (5) 20 (6) $\frac{29}{2}$

2. (1) are collinear. (2) are not collinear. (3) are not collinear. (4) are collinear.

3. (−1, 0) **7.** 7 or −5

Practice set 5.2

1. (1, 3) **2.** (1) $\left(-\frac{1}{3},-\frac{1}{3}\right)$ (2) $\left(\frac{4}{7},-\frac{11}{7}\right)$ (3) $\left(0,\frac{13}{3}\right)$ **3.** 2:7 **4.** (−6, 3)

5. 2:5, $k = 6$ **6.** (11, 18) **7.** (1) (1, 3) (2) (6, −2) (3) $\left(\frac{19}{3},\frac{22}{3}\right)$

8. (−1, −7) **9.** $h = 7$, $k = 18$ **10.** (0, 2) ; (−2, −3)

11. (−9, −8), (−4, −6), (1, −4) **12.** (16, 12), (12, 14), (8, 16), (4, 18)

Practice set 5.3

1. (1) 1 (2) $\sqrt{3}$ (3) slope cannot be determined.

2. (1) 2 (2) $-\frac{3}{8}$ (3) $\frac{5}{2}$ (4) $\frac{5}{4}$ (5) $\frac{1}{2}$ (6) slope cannot be determined.

3. (1) are collinear. (2) are collinear. (3) are not collinear. (4) are collinear.

(5) are collinear. (6) are collinear.

4. −5; $\frac{1}{5}$; $-\frac{2}{3}$ 6. $k = 5$ 7. $k = 0$ 8. $k = 5$

Problem set 5

1. (1) D (2) D (3) C (4) C

2. (1) are collinear. (2) are collinear. (3) are not collinear. 3. (6, 13) 4. 3:1

5. (−7, 0) 6. (1) $a\sqrt{2}$ (2) 13 (3) $5a$ **7.** $\left(-\frac{1}{3}, \frac{2}{3}\right)$

8. (1) Yes, scalene triangle (2) No. (3) Yes, equilateral triangle 9. $k = 5$

13. 5, $2\sqrt{13}$, $\sqrt{37}$ **14.** (1, 3) **16.** $\left(\frac{25}{6}, \frac{13}{6}\right)$, radius = $\frac{13\sqrt{2}}{6}$ **17.** (7, 3)

18. Parallelogram **19.** A(20, 10), P(16, 12), R(8, 16), B(0, 20). **20.** (3, −2)

21. (7, 6) and (3, 6) **22.** 10 and 0

Chapter 6 Trigonometry

Practice set 6.1

1. $\cos\theta = \frac{24}{25}$; $\tan\theta = \frac{7}{24}$ **2.** $\sec\theta = \frac{5}{4}$; $\cos\theta = \frac{4}{5}$

3. $\operatorname{cosec}\theta = \frac{41}{9}$; $\sin\theta = \frac{9}{41}$ **4.** $\sec\theta = \frac{13}{5}$; $\cos\theta = \frac{5}{13}$; $\sin\theta = \frac{12}{13}$

5. $\frac{\sin\theta + \cos\theta}{\sec\theta + \operatorname{cosec}\theta} = \frac{1}{2}$

Practice set 6.2

1. Height of the church is 80 metre.
2. The ship is 51.90 metre away from the ligthhouse.
3. Height of the second building is $(10 + 12\sqrt{3})$ metre.
4. Angle made by the wire with the horizontal line is 30°.
5. Height of the tree is $(40 + 20\sqrt{3})$ metre.
6. The length of the string is 69.20 metre.

Problem set 6

1. (1) A (2) B (3) C (4) A

2. $\cos\theta = \frac{60}{61}$ **3.** $\sin\theta = \frac{2}{\sqrt{5}}$; $\cos\theta = \frac{1}{\sqrt{5}}$; $\operatorname{cosec}\theta = \frac{\sqrt{5}}{2}$; $\sec\theta = \sqrt{5}$; $\cot\theta = \frac{1}{2}$

4. $\sin\theta = \frac{5}{13}$; $\cos\theta = \frac{12}{13}$; $\operatorname{cosec}\theta = \frac{13}{5}$; $\tan\theta = \frac{5}{12}$; $\cot\theta = \frac{12}{5}$

6. Height of the building is $16\sqrt{3}$ metre.

7. The ship is $100\sqrt{3}$ metre away from the ligthhouse.

8. Height of the second buliding is $(12 + 15\sqrt{3})$ metre.

9. The maximum height that ladder can reach is 20.80 metre.

10. the plane was 1026 metre high at the time of landing.

Chapter 7 Mensuration

Practice set 7.1

1. 11.79 cm^3 **2.** 113.04 cm^3 **3.** 1413 sq.cm (by taking $\pi = 3.14$) **4.** 616 sq.cm
5. 21 cm **6.** 12 jugs **7.** 9 cm **8.** 273π sq.cm **9.** 20 tablets
10. 94.20 cm^3, 103.62 sq.cm **11.** 5538.96 sq.cm, 38772.72 cm^3
12. 1468.67π cm^3

Practice set 7.2

1. 10.780 litre **2.** (1) 628 sq.cm (2) 1356.48 sq.cm (3) 1984.48 cm^3

Practice set 7.3

1. 47.1 sq.cm **2.** 25.12 cm **3.** 3.85 sq.cm **4.** 214 sq.cm **5.** 4 cm
6. (1) 154 sq.cm (2) 25.7 sq.cm (3) 128.3 sq.cm **7.** 10.2 sq.cm
8. 7.3 cm ; 22 cm **9.** (1) 90° (2) 22 cm
10.(1) 12.83 sq.cm (2) 89.83 sq.cm (3) 115.5 sq.cm **11.** 3.5 cm
12. x = 154 sq.cm ; y = 38.5 sq.cm ; z = 101.5 sq.cm
13. (1) 84.87 sq.cm (2) 25.67 sq.cm (3) 77.01 sq.cm (4) 7.86 sq.cm

Practice set 7.4

1. 3.92 sq.cm **2.** 9.08 sq.cm **3.** 0.65625 sq.unit **4.** 20 cm
5. 20.43 sq.cm ; 686.07 sq.cm

Problem set 7

1. (1) A, (2) D, (3) B, (4) B, (5) A, (6) A, (7) D, (8) C.
2. 20.35 litre **3.** 7830 balls **4.** 2800 coins (by taking $\pi = \frac{22}{7}$) **5.** Rs. 6336
6. 452.16 sq.cm ; 3385.94 gm **7.** 2640 sq.cm **8.** 243 metre
9. 150°; 5π cm **10.** 39.28 sq.cm

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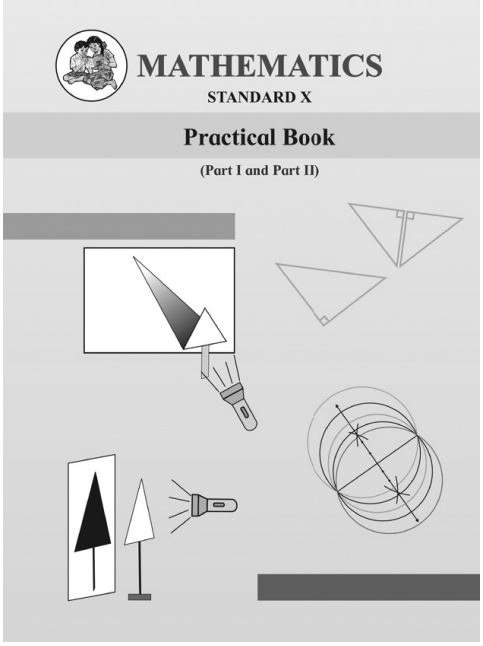

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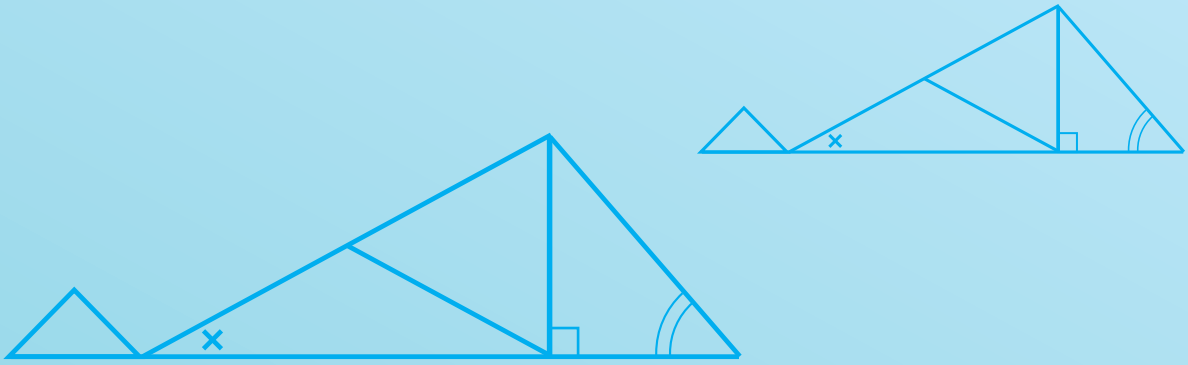

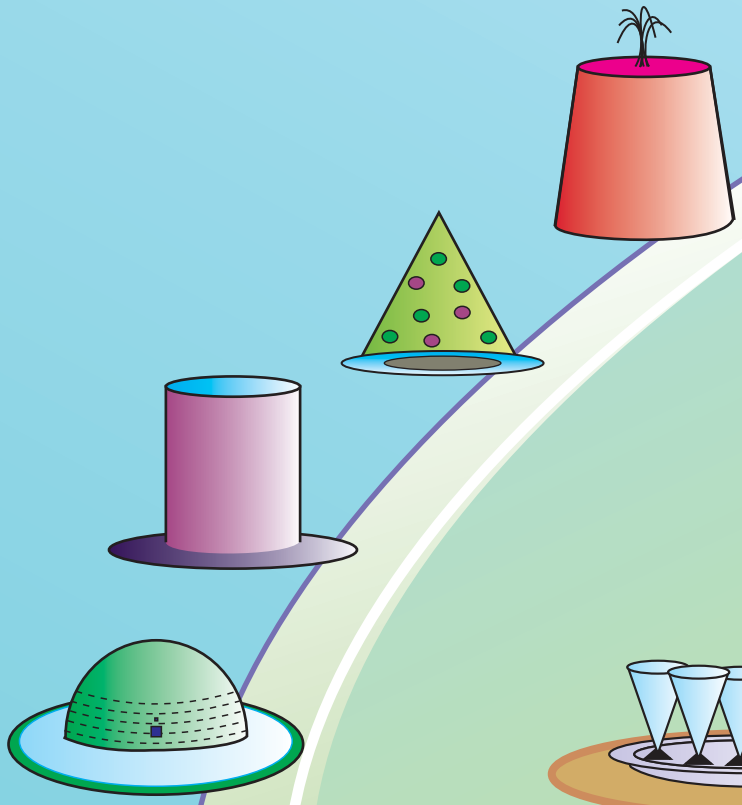

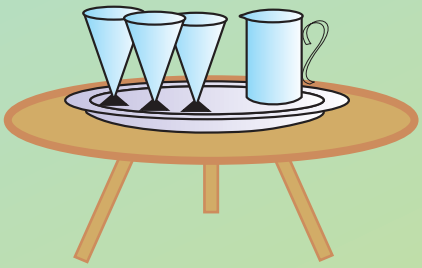

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